

بوابة دفع باستخدام Moyasar

بوابة الدفع هي خدمة إلكترونية تتيح للمستخدمين إجراء المعاملات المالية عبر الإنترنت بأمان وسهولة. تعمل بوابات الدفع كوسيط بين العملاء والتجار، حيث تقوم بمعالجة عمليات الدفع باستخدام بطاقات الائتمان أو المحافظ الرقمية. في هذا المستند، سنوضح كيفية إنشاء بوابة دفع إلكترونية باستخدام مزود خدمة الدفع "Moyasar" ضمن Spring Boot.

إليك الخطوات:

1- إنشاء مشروع Spring Boot

2- إنشاء نموذج الدفع Model

```

5 usages
6 @Data
7 @AllArgsConstructor
8 @NoArgsConstructor
9 public class PaymentRequest {
10     private String name;
11     private String number;
12     private String cvc;
13     private String month;
14     private String year;
15     private double amount;
16     private String currency;
17     private String description;
18     private String callbackUrl;
19
20
21

```

-3 ضبط application.properties :

```

## replace your Moyasar.api.key from your Moyasar account
moyasar.api.key=sk_test_XXXXXXXXXXXXXXXXX

```

-4 إنشاء خدمة للتعامل مع مزود الدفع

```

2 usages
@Service
@RequiredArgsConstructor
public class PaymentService {

    @Value("${moyasar.api.key}") //retrieve it from application.properties
    private String apiKey;
    2 usages
    private static final String MOYASAR_API_URL = "https://api.moyasar.com/v1/payments/";

```

-5- ننشئ دالة إتمام عملية الدفع:

```

1 usage
public ResponseEntity<String> processPayment(PaymentRequest paymentRequest){

    String url = "https://api.moyasar.com/v1/payments";

    // Add your callback URL (replace with your actual callback endpoint)
    String callbackUrl = "https://your-server.com/api/payments/callback";

    // Create the request body
    String requestBody = String.format(
        "source[type]=card&source[name]=%s&source[number]=%s&source[cvc]=%s&" +
        "source[month]=%s&source[year]=%s&amount=%d&currency=%s" +
        "&callback_url=%s",
        paymentRequest.getName(),
        paymentRequest.getNumber(),
        paymentRequest.getCvc(),
        paymentRequest.getMonth(),
        paymentRequest.getYear(),
        (int) (paymentRequest.getAmount() * 100), // Convert to smallest currency unit
        paymentRequest.getCurrency(),
        callbackUrl
    );

```

```

// Set headers
HttpHeaders headers = new HttpHeaders();
headers.setBasicAuth(apiKey, password: "");
headers.setContentType(MediaType.APPLICATION_FORM_URLENCODED);

HttpEntity<String> entity = new HttpEntity<>(requestBody, headers);

// Send the request
RestTemplate restTemplate = new RestTemplate();
ResponseEntity<String> response = restTemplate.exchange(url,
    HttpMethod.POST, entity, String.class);
// Return the API response

return ResponseEntity.status(response.getStatusCode()).body(response.getBody());
}

```

-6 في وحدة التحكم:

ننشئ في طبقة التحكم (Controller Layer) بشكل كائن PaymentRequest

```

@RestController
@RequestMapping("/api/v1/payments")
@RequiredArgsConstructor
public class PaymentController {
    private final PaymentService paymentService;

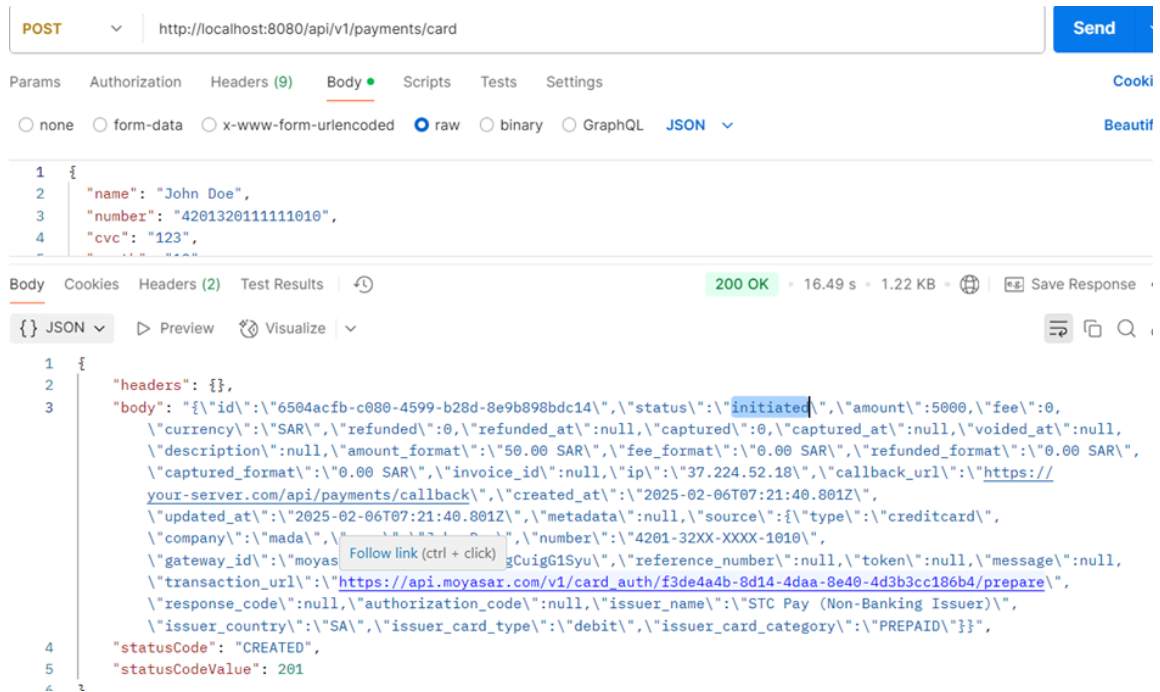
    @PostMapping("/card")
    public ResponseEntity<ResponseEntity<String>>
    processPayment(@RequestBody PaymentRequest paymentRequest) {

        // Return the API response
        return ResponseEntity.status(200).body(paymentService.processPayment(paymentRequest));
    }
}

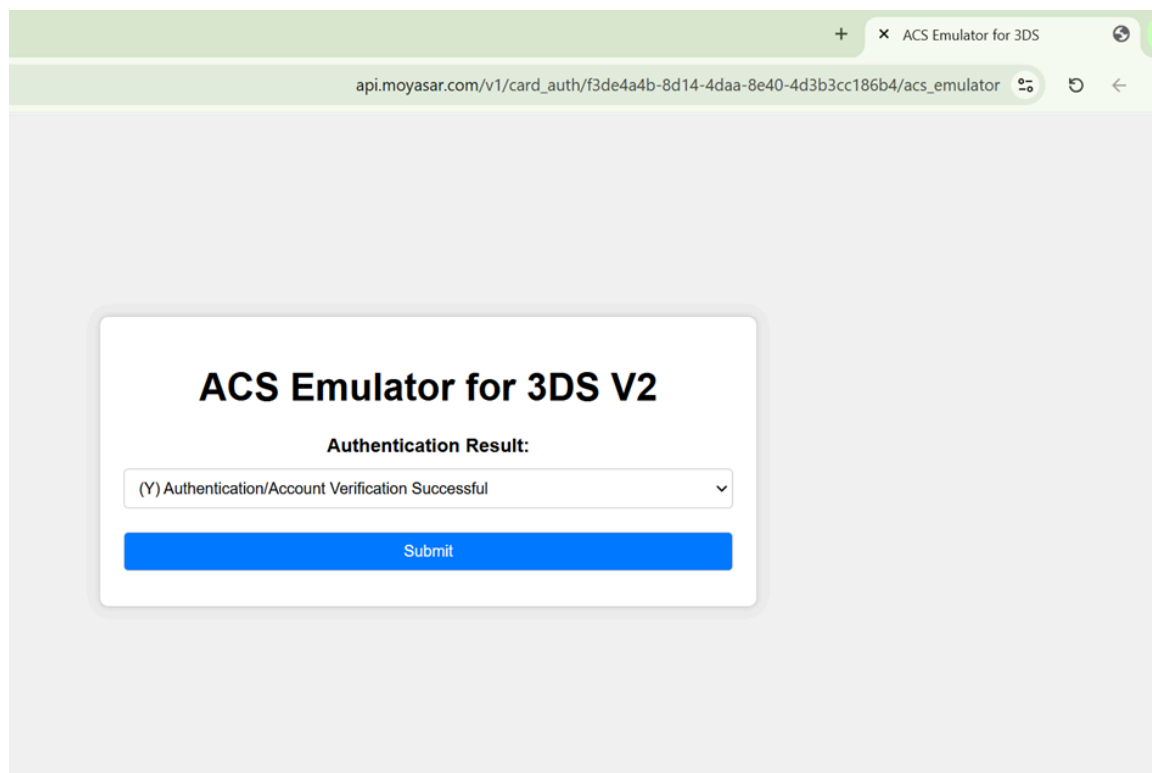
```

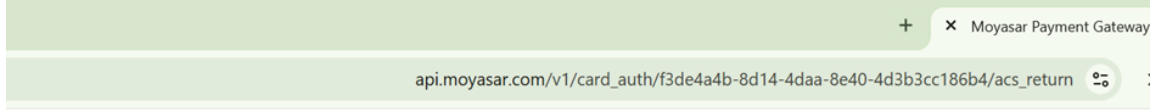
-7 نختبر عملية الدفع من خلال Postman

لاحظ العملية ابتدأت، ولكن لم تكتمل، لذلك لابد من تأكيد العملية من خلال الرابط المرسل لك transaction_url



8- ننسخ رابط transaction_url ونأكد من خلال submit





⌛ You are being redirected back to merchant...



9- في وحدة الخدمة **PaymentService**

نقوم بإنشاء دالة نستعلم حالة الدفع من خلال **id payment** المرسل هنا

POST pa GET gets GET https GET https GET https GET New GET https GET https No environment

Book / pay Save Share

POST http://localhost:8080/api/v1/payments/card Send

Params Authorization Headers (9) Body Scripts Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Beautify

```

1 {
2   "name": "John Doe",
3   "number": "420132011111010",
4   "cvc": "123",

```

Body Cookies Headers (2) Test Results 200 OK 16.49 s 1.22 KB Save Response

{ } JSON Preview Visualize

```

1 {
2   "headers": {},
3   "body": "{ \"id\": \"6504acfb-c080-4599-b28d-8e9b898bdc14\", \"status\": \"initiated\", \"amount\": 5000, \"fee\": 0,
  \"currency\": \"SAR\", \"refunded\": 0, \"refunded_at\": null, \"captured\": 0, \"captured_at\": null, \"voided_at\": null,
  \"description\": null, \"amount_format\": \"50.00 SAR\", \"fee_format\": \"0.00 SAR\", \"refunded_format\": \"0.00 SAR\",
  \"captured_format\": \"0.00 SAR\", \"invoice_id\": null, \"ip\": \"37.224.52.18\", \"callback_url\": \"https://
  your-server.com/api/payments/callback\", \"created_at\": \"2025-02-06T07:21:40.801Z\",
  \"updated_at\": \"2025-02-06T07:21:40.801Z\", \"metadata\": null, \"source\": { \"type\": \"creditcard\",
  \"company\": \"mada\", \"name\": \"John Doe\", \"number\": \"4201-32XX-XXXX-1010\",
  \"gateway_id\": \"moyasar_cc_8rDNA6SE23xP2gCuigG1Syu\", \"reference_number\": null, \"token\": null, \"message\": null,
  \"transaction_url\": \"https://api.moyasar.com/v1/card_auth/f3de4a4b-8d14-4daa-8e40-4d3b3cc186b4/prepare\",
  \"response_code\": null, \"authorization_code\": null, \"issuer_name\": \"STC Pay (Non-Banking Issuer)\",
  \"issuer_country\": \"SA\", \"issuer_card_type\": \"debit\", \"issuer_card_category\": \"PREPAID\" } }",
4   "statusCode": "CREATED",
5   "statusCodeValue": 201
6 }

```

الدالة تستقبل paymentId

```

1 usage
public String getPaymentStatus(String paymentId) {
    // Prepare headers with authentication
    HttpHeaders headers = new HttpHeaders();
    headers.setBasicAuth(apiKey, password: ""); // Basic Auth (API Key as username)
    headers.setContentType(MediaType.APPLICATION_JSON);

    // Create HTTP request entity
    HttpEntity<String> entity = new HttpEntity<>(headers);

    // Call Moyasar API
    RestTemplate restTemplate = new RestTemplate();
    ResponseEntity<String> response = restTemplate.exchange(
        url: MOYASAR_API_URL + paymentId, HttpMethod.GET, entity, String.class
    );

    // Return the response
    return response.getBody();
}

```

10- في وحدة التحكم:

ننشئ في طبقة التحكم (Controller Layer)

```
@GetMapping(⊕"get-status/{id}")
public ResponseEntity getPaymentStatus(@PathVariable String id){
    return ResponseEntity.status(HttpStatus.OK).body(paymentService.getPaymentStatus(id));
}
}
```

11- تستعلم عن عملية الدفع من خلال Postman

GET http://localhost:8080/api/v1/payments/get-status/6504acfb-c080-4599-b28d-8e9b898bdc14 Send

Params Authorization Headers (7) Body Scripts Tests Settings

Cookies

☒ none ☐ form-data ☐ x-www-form-urlencoded ☐ raw ☐ binary ☐ GraphQL

This request does not have a body

Body Cookies Headers (2) Test Results 200 OK 7.46 s 995 B Save Response

Raw Preview Visualize

```
1 {"id":"6504acfb-c080-4599-b28d-8e9b898bdc14","status":"paid","amount":5000,"fee":0,"currency":"SAR","refunded":0,"refunded_at":null,"captured":0,"captured_at":null,"voided_at":null,"description":null,"amount_format":"50.00 SAR","fee_format":"0.00 SAR","refunded_format":"0.00 SAR","captured_format":"0.00 SAR","invoice_id":null,"ip":"37.224.52.18","callback_url":"https://your-server.com/api/payments/callback","created_at":"2025-02-06T07:21:40.801Z","updated_at":"2025-02-06T07:28:12.220Z","metadata":null,"source":{"type":"creditcard","company":"mada","name":"John Doe","number":"4201-32XX-XXXX-1010","gateway_id":"moyasar_cc_8rDNA6SE23xP2gCuigG1Syu","reference_number":"992503759510","token":null,"message":"APPROVED","transaction_url":null,"response_code":"00","authorization_code":"004777","issuer_name":"MOYASAR PAY SANDBOX","issuer_country":"SA","issuer_card_type":"credit","issuer_card_category":"PREPAID"}}
```

في حالة ذهبت لحسابك الشخصي في Moyasar سوف ترى حالات الدفع التي جرت من خلال
:YOUR SECRET API و Moyasar API

Student
Switch Account

Services

Payments

Invoices

Configuration

Team

Settings

Not Active

Help

Payments

Export

Payment ID + Status + Source + Metadata + Last 4 digits +

Sort Clear All

Reference + Created at + Updated at +

STATUS	SOURCE	PAYMENT ID	CURRENCY	AMOUNT	DESCRIPTION	CREATED AT
PAID	CS20 made	8e9b898bdc14	SAR	50.00	-	06/02/2025 - 10:21 AM
PAID	CS20 made	fc3258138892	SAR	50.00	-	03/02/2025 - 11:02 AM
FAILED	CS20 made	d0ccf48b1524	SAR	50.00	-	03/02/2025 - 11:00 AM
FAILED	CS20 made	7e4d9e1c424c	SAR	50.00	-	03/02/2025 - 10:56 AM